

Lesson 5.



$$y = x\beta + u \quad u \sim N(0, \sigma^2 I)$$

$$H_0: R\beta = r$$

$q \times k \quad k \times 1 \quad \rightarrow \quad q \times 1$

Vigression: Let $z \sim N(0, I_n)$

$n \times 1$

$$\sum_{i=1}^n z_i^2 = z^T z \sim \chi_n^2$$

$1 \times n \quad n \times 1$

$$\text{If } x \sim N(0, \bar{\Sigma}) \quad x^T \bar{\Sigma}^{-1} x \sim \chi_n^2$$

$n \times 1$

Spectral decomposition The'm

Let $\bar{\Sigma}$ be symmetric and real

$$\text{then } \bar{\Sigma} = \underbrace{\Lambda}_{n \times n} \underbrace{Q}_{\text{scalar}}$$

$$Q^T \bar{\Sigma} Q = \Lambda \quad \Lambda = \begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix} \quad \text{here } Q^T = Q^{-1} \rightarrow \text{orthogonal matrix.}$$

$$Q Q^T \bar{\Sigma} Q Q^T = Q \Lambda Q^T$$

$$\bar{\Sigma} = Q \Lambda Q^T \rightarrow Q \Lambda^{\frac{1}{2}} \Lambda^{\frac{1}{2}} Q^T \quad \text{where } \Lambda^{\frac{1}{2}} = \begin{bmatrix} \sqrt{\lambda_1} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sqrt{\lambda_n} \end{bmatrix}$$

$$Q \Lambda^{\frac{1}{2}} \Lambda^{\frac{1}{2}} Q^T = \bar{\Sigma}$$

$$T T' = \bar{\Sigma}$$

$$z \sim N(0, I_n) \quad x = Tz$$

$$n \times 1$$

$$E(x) = T E(z) = 0$$

$$\text{Var}(x) = T \text{Var}(z) T' = T T' = \bar{\Sigma}$$

$$z = T^{-1} x \quad z^T z \sim \chi_n^2$$

$$x^T (T^{-1})' T^{-1} z = x^T (x')^{-1} T^{-1} x = x^T \bar{\Sigma}^{-1} x \sim \chi_1^2$$

$$\bar{\Sigma}^{-1} = (T^{-1})' (T^{-1})$$

(3) Let $z \sim N(0, I_n)$

Let A be idempotent of rank p . Then $z^T A z \sim \chi_p^2$

$$A \times A = A$$

$$A \times q = \lambda q$$

$$A(Aq) = A \lambda q \quad Aq = \lambda Aq = \lambda^2 q$$

$$\lambda q = X'q = (X'X - X)q = 0$$

$$\lambda = 1 \text{ or } \lambda = 0$$

$$\text{rank}(A) = \text{trace}(A)$$

$$Q^T A Q = \Lambda \quad \text{trace}(Q^T A Q) = \text{trace}(\Lambda) \Rightarrow \sum_{i=1}^n \lambda_i = P$$

$$\text{trace}(Q^T A Q) = \text{trace}(\Lambda) = P = \text{trace}(A) = P$$

$$Z = QY \quad Y = Q^T Z \quad E(Y) = 0 \quad \text{Var}(Y) = Q^T \text{Var}(Z) = Q^T Q = I_n$$

$$Y'Y \sim \chi^2_P \quad Z^T A Z = Y' Q^T A Q Y \xrightarrow{\lambda} = Y' \Lambda Y = \sum_{i=1}^P y_i^2 \sim \chi^2_P$$

(4) Let A_X and B_X be two normal vector where A and B are idempotent, symmetric of P .

and q ranks respectively.

$$w_1 = z' A z \sim \chi_p^2 \quad w_2 = z' B z \sim \chi_q^2$$

w_1 is independent of w_2 iff $AB = 0$

$$\begin{matrix} A_z & B_z \\ E(A_z (B_z)') = A \text{Var}(z) B = AB \\ & = 0 \end{matrix}$$

Definition Let $w_1 \sim \chi_p^2$ and $w_2 \sim \chi_q^2$

$$\frac{w_1/p}{w_2/q} \sim F(p, q)$$

$$w_1 \sim \chi_1^2 \Rightarrow \frac{w_1/1}{w_2/q} = F(1, q) = t_q^2$$

$$H_0: RB = r$$

$$\begin{aligned} & |Rb - r| \quad X \bar{X}' X \\ \hookrightarrow & (Rb - r)' (\text{Var}(Rb - r)) (Rb - r) \sim \chi_q^2 \end{aligned}$$

$$Rb - r = R(X'X)^{-1} X' y - r$$

$$= RB + R(X'X)^{-1} X' u - r$$

$$= R(X'X)^{-1} X' u \quad \text{when } H_0$$

$$\text{Var}(Rb-r) \stackrel{\text{Var}}{=} (R(X'X)^{-1} X' u)$$

$$= R(X'X)^{-1} \underbrace{E(uu^T)}_{\sigma^2 I} X(X'X)^{-1} R$$

$$= \sigma^2 (R(X'X)^{-1} R')$$

sandwich

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$$\text{Var}(AX)$$

$$= A \text{Var}(X) A'$$

$$|Rb-r| = (Rb-r)' (\text{Var}(Rb-r)^{-1}) (Rb-r)$$

$$Rb-r = R(X'X)^{-1} X' u$$

$$\sigma^2 u' \underbrace{X(X'X)^{-1} R (R(X'X)^{-1} R')^{-1} R(X'X)^{-1} X'}_N u$$

$$= u' N u$$

$$m N = m X (---)$$

$$N' = N \quad N \times N = N$$

$$\text{rank}(N) = \text{trace}(N)$$

$$\text{trace} \left[(X(X'X)^{-1}R') (R(X'X)^{-1}R')^{-1} R(X'X)^{-1}X' \right]$$

$$= \text{trace} \left(R(X'X)^{-1}R \right)^{-1} \left(R(X'X)^{-1}X'X(X'X)^{-1}R' \right)$$

$$= \text{trace}(I_q) = q$$

$$\zeta^2 = \frac{u'mu}{n-k}$$

$$\textcircled{1} u'Nu \sim \chi^2_q \quad \textcircled{2} u'Mu \sim \chi^2_{n-k}$$

$MN=0$ because $MX=0$

$$\frac{\frac{1}{b^2} u' N' u \mid q}{\frac{1}{b^2} u' M u \mid n-k} \rightarrow F(q, n-k)$$

$\frac{1}{b^2} u' N' u \mid q \rightarrow \chi^2_q$
 $\frac{1}{b^2} u' M u \mid n-k \rightarrow \chi^2_{n-k}$